# The Art of Communicating Science - A Guide & Checklist

## LS100 — Module 03 · Communication & Presentation

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Science communication is not about translating scientific knowledge into simplified take-home messages; it is about connecting with and evoking thoughts about scientific topics in the audience through narrative storytelling. Effective science communication builds understanding, curiosity, and trust in science by linking evidence with human experience. This article outlines a framework for designing and delivering oral presentations that combine narrative structure, visual clarity, and authentic delivery. The key concepts include audience mapping, narrative framing, visual design, and delivery of the talk - emphasizing coherent storytelling without diluting scientific rigor. Although the focus on this article is on oral presentation, the underlying principles—contextual framing, ethical data representation, and emotional resonance—extend to other formats of science communication including posters, written formats such as popular articles or research papers, and audiovisual media. The article concludes with practical checklists and rubrics that support both preparation and reflective evaluation of scientific communication practice.

### *0.a. Introduction*

Science communication is not primarily a matter of translating complex facts into simplified slogans. It is the practice of **connecting human concerns to evidence** in ways that help audiences make sense of questions that matter to them – e.g., knowledge production, applied sciences like health, environment, technology, and public life. Doing this requires aligning **what the audience cares about** with **what the evidence shows** and **what decisions or actions are at stake**. Research syntheses from the National Academies emphasize that effective practice is audience- and context-dependent, and that communicators should move beyond the “deficit” assumption (more facts arrow.r more agreement) toward approaches that consider values, identities, and goals [1].

Growing evidence base supports **narrative** as a particularly potent structure for that alignment. Narrative formats can increase comprehension, interest, and engagement—especially for non-expert audiences who encounter most science via narrative-biased media—provided stories are used ethically and do not distort uncertainty or mechanisms. In science settings, narrative should serve accuracy: a



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clear arc (problem arrow.r approach arrow.r evidence arrow.r implications) accompanied by transparent limits [Dahlstrom (2014)].

Because oral presentations frequently combine **spoken words with visuals**, communicators benefit from principles in multimedia learning and slide design. Decades of research (e.g., Mayer's cognitive theory) recommend reducing **extraneous load** (coherence), guiding attention (signaling), and placing words and graphics together (contiguity) to improve understanding. These principles generalize well to research talks, classes, and public lectures [3].

Within that multimedia context, the **assertion–evidence** slide model has shown advantages over default "topic-headline + bulleted text" slides. In controlled comparisons, assertion–evidence decks (claim-title + visual support + minimal text) improved comprehension and reduced perceived cognitive load for technical audiences, and programmatic resources now exist to help scientists adopt the approach [4].

Finally, **engagement is part of rigor**. Brief interactions (questions, prompts, checks for understanding) make talks more than one-way transmission; they help audiences process and apply ideas. Meta-analytic evidence from STEM teaching shows that active engagement (vs. pure lecturing) yields higher performance and lower failure rates, a signal that practices which invite audience thinking are pedagogically sound even in expert settings. While research talks are not courses, the same cognitive and motivational mechanisms apply when audiences are asked to reason with evidence rather than simply receive it [Freeman et al. (2014)].

This article takes a **reflective-pedagogy** stance: it distills what practitioners can do before, during, and after an oral presentation so that narrative, visuals, and delivery reinforce one another without compromising scientific integrity. Although our focus is oral communication, the underlying principles—**contextual framing, ethical representation of uncertainty, and emotional resonance disciplined by evidence**—readily adapt to posters, writing, and audiovisual work; later sections make those bridges explicit [1].

**Table 1.** Narrative beats: blockbuster films vs. oral science talks

| Narrative Beat | Blockbuster Films (what happens) | Oral Science Talk (what you do) | Example |
|---|---|---|---|
| **Setup / World** | Establish the world and stakes | Frame the *real-world problem* and why it matters | "What decision or outcome is at stake for this audience?" |

| **Narrative Beat** | **Blockbuster Films (what happens)** | **Oral Science Talk (what you do)** | **Example** |
|---|---|---|---|
| **Protagonists** | Introduce characters and goals | Introduce *actors*: system, variables, stakeholders; your research goal | "Whose perspective clarifies the stakes (patient, public, researcher, policy maker, animal, cell, molecule)?" |
| **Known terrain** | Reveal what's already known | Summarize consensus and prior evidence succinctly | "What are the two things the field already agrees on?" |
| **Tension / Unknown** | Raise the central conflict | Name the **specific gap or question** | "What exactly don't we know that blocks progress?" |
| **Plan** | Protagonists choose a path | Explain the **design logic**—why this approach counts as evidence | "If X is true, what should we observe?" |
| **Trials** | Obstacles and attempts | Show key experiments or analyses, including dead ends where informative | "What did we learn from what *didn't* work?" |
| **Resolution** | Conflict resolved | Present **results arrow.r plain-language meaning** with uncertainty | "How big is the effect? How certain are we?" |
| **Aftermath** | Consequences for the world | Zoom out: **implications, limits, next steps, or call to action** | "Given these limits, what should change now?" |

Notes: Use narrative to organize reasoning, not to oversell. Keep fidelity high, avoid metaphors that misrepresent mechanisms and show uncertainty both visually and verbally.

**Table 2.** Suggested structures by purpose and audience

| Purpose / Context | Primary Audience | Recommended Arc (Beats) | Time Budget (12–15 min) | Time Budget (30–45 min) | Notes |
|---|---|---|---|---|---|
| **Research plan / proposal** | Domain peers, funders | Hook arrow.r So-what arrow.r Gap arrow.r **Design logic** arrow.r Proposed studies arrow.r Risks & mitigations arrow.r Impact / ethics arrow.r Ask | 2 / 2 / 2 / 4 / 3 | 4 / 5 / 5 / 10 / 6 | Lead with stakes; foreground *why this approach will produce evidence.* |
| **Completed single study** | Peers across fields | Hook arrow.r So-what arrow.r Known arrow.r Gap arrow.r Methods (conceptual) arrow.r Results arrow.r Meaning + **uncertainty** arrow.r Limits arrow.r Implications | 3 / 2 / 1 / 1 / 2 / 3 / 2 | 6 / 4 / 3 / 3 / 6 / 8 / 4 | Keep methods conceptual; emphasize effect sizes and uncertainty. |

| Purpose / Context | Primary Audience | Recommended Arc (Beats) | Time Budget (12–15 min) | Time Budget (30–45 min) | Notes |
|---|---|---|---|---|---|
| **Program update / progress** | Team, stakeholders | Hook arrow.r Objectives arrow.r Milestones arrow.r What worked / learned arrow.r Risks / needs arrow.r Next steps | 2 / 3 / 4 / 3 / 2 | 5 / 7 / 10 / 5 / 3 | Highlight learning from failures; end with concrete next actions. |
| **Public or policy briefing** | Non-experts, decision-makers | Relatable hook arrow.r So-what (stakes) arrow.r What's known arrow.r What's uncertain arrow.r Implications / options arrow.r Call to action | 3 / 3 / 2 / 2 / 3 | 6 / 6 / 4 / 4 / 6 | Avoid deficit framing; connect to shared values and real-world choices. |

| Purpose / Context | Primary Audience | Recommended Arc (Beats) | Time Budget (12–15 min) | Time Budget (30–45 min) | Notes |
|---|---|---|---|---|---|
| **Investor / partner pitch** | Funders, partners | Problem size arrow.r Solution concept arrow.r Evidence so far arrow.r Path to scale arrow.r Risks arrow.r Ask | 3 / 3 / 4 / 3 / 2 | 6 / 6 / 10 / 7 / 3 | Focus on evidence quality; be explicit about assumptions and risk. |

Note: Time budgets are indicative; adjust to venue and audience. For all versions, use assertion–evidence slides and multimedia principles to reduce extraneous load and guide attention [4].

*0.b. Forms of Science Communication*

While this guide focuses on oral presentations with slides, science communication also takes the form of posters, written work (popular or academic), and audiovisual media such as documentaries. Though the medium changes, shared principles endure: know your audience, tell a story, and make complexity legible.

3. 1. Planning Your Talk

    3. Know your audience: Start by mapping what your audience knows and cares about.
    4. Content of your talk:
    5. Structure of your talk:

Translate this into a Message Box: Problem arrow.r So what arrow.r What we know arrow.r What's new arrow.r Call to action. Then build an hourglass structure: start broad (why it matters), narrow to methods and findings, and end broad again (implications and limits).

*1.a. 4. Designing Slides*

Slides should amplify your story, not replace it. Use assertion–evidence design: each slide title states the takeaway message, and the body shows evidence supporting it. Limit content to one main idea per slide, keep text concise, and favor clear annotations over legends.

*1.b. 5. Delivering the Talk*

You are the medium. Maintain eye contact, vary tone and pacing, and use micro-pauses (1–3 seconds) for emphasis. Engage with your audience through brief questions or prompts. Confidence grows from preparation and empathy, not from volume or speed.

*1.c. 6. Handling Questions*

Listen fully, paraphrase the question for clarity, answer concisely, and bridge back to your message. It's acceptable to say 'I don't know'—follow up with how you'd find the answer. Acknowledge limits and credit collaborators openly.

*1.d. 7. Practice and Logistics*

Rehearse out loud at least once. Record yourself and fix one thing at a time—timing, transitions, filler words. Check your technology, adapters, and accessibility (captions, readable text). Confidence is preparation made visible.

*1.e. 8. Quick Checklist*

- Audience map and one-sentence promise completed.
- Message Box drafted; hourglass structure planned.
- Slides use assertion–evidence format; one idea per slide.
- Figures annotated; uncertainty shown where relevant.
- Interaction planned every ~5 minutes.
- Q&A approach practiced; collaborators credited.
- Backup PDF prepared; accessibility verified.

*1.f. 9. Self-Evaluation Rubric*

Use this rubric (1–5 scale) for self- or peer-assessment:
1 = needs work, 3 = competent, 5 = excellent.
Criteria include audience focus, story structure, slide design, data clarity, delivery, ethics, and rehearsal quality.

*1.g. 10. Adapting to Posters, Writing, and Audio–Visual*

Posters [P]: Use modular sections with headline-style titles that tell a story even if the viewer never meets you. Writing [W]: Use clear topic sentences, signposting, and paragraph-level logic. Audio–Visual [AV]: Plan with a storyboard, write a timed script, and ensure good sound and captioning. The same story spine—Problem, Approach, Findings, Implications—applies to all formats.

## References:

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